

Micro Protection Fluid

Product Proposal — HSBC Life / HSBC General Insurance

Standalone Parametric Insurance | HKD 200/month | Six Coverages

Micro Protection Fluid is a standalone, pure parametric insurance product.

Feature	Specification
Monthly premium	HKD 200 (single plan)
Coverage categories	6 (Life, Health, Travel, Liability, Cyber, Property)
How it works	Rules-based engine reallocates coverage in real-time based on your activity
Claims	Pure parametric, triggered by APIs, no loss verification
Customer view	One product, one app, one premium, one claim

The problem: Traditional insurance charges the same premium whether you're commuting to Shenzhen or sitting in Hong Kong. GBA cross-border commuters are over-covered for static risks and under-covered for dynamic ones.

The opportunity: 15.03 million GBA cross-border trips in H1 2025. 67% of GBA customers under 35 prefer modular, on-demand insurance over bundled life policies.

Our validation: State Farm patented this model. Peak3 has live accumulator insurance in Asia. Swiss Re validates real-time adaptive coverage as the industry's future.

Part 1: The Product (Customer View)

What the Customer Sees

One product. HKD 200/month. Six coverages. All in the HSBC app.

Category	What's Covered	How It Triggers
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Life	Death benefit	Digital death certificate
Health	Hospital admission	Hospital admission API
Travel	Flight delay (>4h), Trip cancellation/interruption, Baggage loss, Travel delays	Flight status API, MTR Corp API, China Customs API
Liability	Third-party claims (supplementary only)	Validated third-party claim
Cyber	Identity theft recovery	Identity theft report
Property	Theft of personal belongings	Police report API

How Allocation Works (Real-Time)

The life allocation of HKD 90 is fixed and permanent; it never participates in real-time reallocation. Only the remaining HKD 110 (the short-term buffer) is dynamically reallocated across health, travel, liability, cyber, and property based on live activity signals.

Parameter	Specification
Monthly premium	HKD 200 fixed
Life allocation	HKD 90 fixed (minimum and maximum — life is always HKD 90, never reallocated)
Minimum per non-life category	HKD 5 (always some coverage)
Maximum per non-life category	HKD 90 (no single risk consumes the pool)
Reallocation frequency	Real-time (sub-second) — non-life categories only
What triggers an increase	MCC codes (airlines, hotels, hospitals), flight booking, geolocation (border crossing), Octopus tap
What triggers a decrease	Same signals indicating risk reduction (return home, flight landed)
Who sets the rules	HSBC Life actuarial team

AI role	Executes within guardrails; suggests optimizations with human approval
What AI cannot do	Create new allocation rules autonomously; adjust Life allocation

The Multiplier Table (Parametric Formula)

Note: Life allocation is fixed at HKD 90, and the minimum and maximum are identical. Life coverage is therefore fixed at HKD 45,000. Only non-life categories participate in dynamic reallocation.

Category	Multiplier	Min Allocation	Min Coverage	Max Allocation	Max Coverage
Life	1:500	HKD 90 (fixed)	HKD 45,000 (fixed)	HKD 90 (fixed)	HKD 45,000 (fixed)
Health	1:200	HKD 5	HKD 1,000	HKD 90	HKD 18,000
Travel	1:150	HKD 5	HKD 750	HKD 90	HKD 13,500
Liability	1:1,000	HKD 5	HKD 5,000	HKD 90	HKD 90,000
Cyber	1:200	HKD 5	HKD 1,000	HKD 90	HKD 18,000
Property	1:100	HKD 5	HKD 500	HKD 90	HKD 9,000

Payout formula: Actual payout = MIN(actual loss, allocation × multiplier)

Important Disclosure (Liability): Liability coverage under Micro Protection Fluid is supplementary only. It is not a substitute for standalone liability insurance. Per-event cap applies. Customers with significant liability exposure should maintain separate coverage.

The "45/55 Life Lock"

Of your HKD 200 monthly premium, the split is fixed as follows:

Pool	Amount	Use
Long-term core reserve	HKD 90 (45%) , fixed, never reallocated	Life claims only, legally segregated. Invested in long-term instruments via HSBC Life.
Short-term liquidity buffer	HKD 110 (55%)	Health, travel, liability, cyber, and property claims. Also provides an early-claim liquidity bridge for life claims before the long-term pool matures.

Legal segregation: The HKD 90 life pool is firewalled by HKIA GL21 and the Insurance Ordinance Cap. 41. No circumstance, such as a surge in short-term claims, a catastrophic event, or AI logic failure, can redirect life funds to non-life categories. The HKD 110 short-term buffer operates independently and also provides an early-claim liquidity bridge for life claims before the long-term pool matures.

Part 2: Regulatory Note

Two-Policy Bundle Structure

Because HSBC Life has a long-term license (not general insurance), the product is internally structured as two policies issued by two separate licensed entities:

Component	Issuing Entity	License
Life coverage	HSBC Life (International) Limited	Long-term insurance (existing)
Health, Liability, Property	Travel, Cyber, HSBC General Insurance Company Limited	General insurance (existing, separate entity)

Customer experience: Single premium (HKD 200), single app, single claim. HSBC internal settlement handles the split. The customer never sees this structure, they see one product called Micro Protection Fluid.

Premium Collection & Inter-Entity Settlement Architecture

The HKD 200 monthly premium is collected through a single customer-facing transaction via the HSBC app. Internally, it is routed through a designated Premium Settlement Account (PSA), an HKIA-compliant intermediary account operated by The Hongkong and Shanghai Banking Corporation Limited in its capacity as licensed insurance agent for both HSBC Life (International) Limited and HSBC General Insurance Company Limited.

The PSA carries no underwriting function and holds no insurance risk. Its sole purpose is premium routing and inter-entity settlement.

Step	Action	Amount	Timeline
1	Customer pays via the HSBC app	HKD 200	T
2	Full premium lands in Premium Settlement Account (PSA)	HKD 200	T
3	Swept to HSBC Life — long-term life reserve	HKD 90 (45%)	T+1
4	Swept to HSBC General Insurance — short-term buffer	HKD 110 (55%)	T+1
5	Daily reconciliation filed internally	—	T+1
6	Monthly reporting to HKIA	—	Monthly

Legal characterisation of the PSA:

- Not an asset of HSBC, the bank
- Not available to HSBC's general creditors under any circumstance

- Held on trust for HSBC Life policyholders (HKD 90 per premium) and HSBC General Insurance policyholders (HKD 110 per premium)
- Governed by a documented tripartite agency agreement between HSBC (as collecting agent), HSBC Life, and HSBC General Insurance

Solvency independence:

Each entity's solvency margin is calculated independently under the Insurance Ordinance Cap. 41. HSBC Life's solvency calculation includes only premium settled to its own account. HSBC General Insurance's solvency calculation includes only premium settled to its own account. The PSA balance is excluded from both entities' solvency calculations at all times.

Regulatory precedent:

This structure is not novel. HSBC currently operates as a licensed insurance agent for both HSBC Life and AXA General Insurance Hong Kong, collecting premiums on behalf of both underwriting entities through unified HSBC banking channels. The PSA for Micro Protection Fluid applies the same legal architecture to an internally structured product, with one addition: a contractually fixed allocation ratio (45% / 55%) governed by the tripartite agency agreement.

GBA Cross-Border Claims

Leg	Mechanism
Mainland payout	HSBC Life China + HSBC General Insurance China (CBIRC licenses)
HK settlement	HSBC Hong Kong
Cross-border settlement	mBridge (BIS Innovation Hub / HKMA / PBOC) — HSBC is a participant

Regulatory Approvals Required

Approval	Authority	Estimated Timeline
Parametric product filing	HKIA	3–4 months
Two-policy coordination bundle	HKIA (Life + GI)	2–3 months

PSA designation and tripartite agency agreement	HKIA	1–2 months
GBA sandbox participation	CBIRC	3–6 months
mBridge integration	HKMA	Already live

Estimated total timeline to launch: 12–14 months

Part 3: Why Standalone (why not as an add-on)?

The Segment We Target

GBA commuters ages 22–35. Cross-border workers. Gig economy participants. Frequent travelers. This segment actively rejects traditional life insurance.

Source	Finding
Swiss Re Institute (2024)	67% of GBA customers under 35 prefer modular, on-demand insurance over bundled life policies
McKinsey GBA Survey (2025)	Only 23% of cross-border commuters own long-term life insurance; 78% want short-term, activity-linked coverage
Peak3 user data (2024)	SNACK accumulator saw 3x higher adoption when offered separately from traditional life
HSBC internal estimate	Gen Z GBA customers have <15% cross-sell from banking to life insurance

Conclusion: Bundling as a life add-on would reduce adoption by an estimated 50–60%. Standalone positioning captures this untapped market without cannibalizing existing life customers.

Part 4: Financial Projections

Key Assumptions

Assumption	Value	Justification
Monthly premium	HKD 200	Fixed, single plan
Life reserve (fixed)	HKD 90 (45%)	Fixed allocation; invested long-term; never reallocated to other categories
Short-term buffer	HKD 110 (55%)	Covers non-life claims; also bridges early life claims pending long-term pool maturity
Loss ratio	55%	Pure parametric, no fraud leakage
Expense ratio (Year 1)	100% of premium	Pilot inefficiency
Expense ratio (Year 2)	40%	Scale economies begin
Expense ratio (Year 3)	35%	Mature operations
Reinsurance cession	15%	Catastrophic coverage only
Customer acquisition	HKD 0 marginal	HSBC app cross-sell to existing customers

Three-Year Forecast

Line Item	Year 1	Year 2	Year 3
Customers (end of year)	12,500	75,000	250,000
Gross premium	HKD 30M	HKD 180M	HKD 600M
Less: Reinsurance (15%)	HKD 4.5M	HKD 27M	HKD 90M
Net premium	HKD 25.5M	HKD 153M	HKD 510M
Less: Parametric claims (55%)	HKD 14.0M	HKD 84.2M	HKD 281M

Less: Operating expenses	HKD 25.5M (100%)	HKD 61.2M (40%)	HKD 179M (35%)
Profit (Loss) before tax	HKD -14M	HKD 7.6M	HKD 50M

Part 5: Validation & Market Data

Real-World Validation

Reference	What It Proves
State Farm patent US20210166322A1	Dynamically reconfigurable insurance is patent-validated
Peak3 accumulator (Singapore, live)	Micro-policies triggered by daily activity work commercially
Swiss Re sigma report 1/2020	Real-time adaptive coverage is an industry-validated future
Singlife / Aviva Singapore merger	Life + general insurance unified under one customer app via a separate-entity bundle — direct structural precedent
FWD Hong Kong (DIHC structure)	HKIA group-wide supervision already accommodates life + general insurance entities operating under one brand

GBA Market Opportunity

Metric	Value	Source
GBA cross-border trips (H1 2025)	15.03M	CCTV / People's Daily
Daily average	83,000	CCTV / People's Daily
YoY growth	16.1%	CCTV / People's Daily
Global protection gap	USD 9T	Swiss Re Institute

Uninsured gig workers (Asia)	70%	ILO 2024
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Part 6: Risk & Mitigation

Risk	Impact	Mitigation	Owner
Regulatory: composite license	High	Two-policy bundle (HSBC Life + HSBC GI)	Legal
Regulatory: PSA commingling	Medium	Tripartite agency agreement; T+1 settlement; trust deed; daily reconciliation	Legal / Treasury
Regulatory: cross-border claims	High	mBridge + HSBC Life/HSBC GI China	Legal
Regulatory: dynamic allocation	Medium	File as rules-based parametric; pre-approved floors/ceilings; life allocation is fixed and excluded from reallocation engine	Compliance
Operational: API failures	Medium	Fallback to baseline coverage	Engineering
Privacy: data misuse	High	Granular opt-in consent (PDPO DPP3 compliant)	Legal
Customer: AI skepticism	Medium	Human-in-the-loop; explainable rules; opt-out	Product
Financial: liability tail risk	Medium	Per-event cap; bold disclosure; reinsurance	Actuarial
Financial: PSA settlement delay	Medium	Automated sweep with manual override; HKIA notified if settlement exceeds T+2	Treasury

Financial: liquidity	Low	Tiered buffer; LCR >200%	Treasury
Financial: adoption	Medium	HKD 200 price point optimized for 75% acceptance	Product
Financial: lower profit vs. add-on	Low	Accepted trade-off for market access and brand independence	Strategy

Part 7: Roadmap

Phase	Timeline	Milestones
Phase 1: Regulatory & Partnerships	Months 1–6	Two-policy bundle agreement (HSBC Life + HSBC GI); HKIA parametric filing; PSA tripartite agency agreement; GBA sandbox application
Phase 2: Build	Months 4–10	Rules engine development; API integration (MCC, flight, Octopus, mBridge); PSA payment rail build
Phase 3: Pilot (HK only)	Months 10–14	10,000 HSBC retail customers; manual guardrails; live monitoring; T+1 settlement validation
Phase 4: GBA Launch	Months 14–18	Expand to GBA; full parametric triggers
Phase 5: Scale	Months 18–24	AI optimization (human-approved); customer growth to 250k; target real-time PSA settlement

Summary: The customer pays one premium. Regulatory compliance is achieved not by merging two licensed entities, but by routing that premium through an HKIA-designated agent account that settles to each underwriting entity within 24 hours — a structure HSBC already operates in its existing bancassurance business.

Part 8: ESG & Sustainability Framework

Micro Protection Fluid is not incidentally ESG-aligned — it is structurally built around the three pillars. The product's parametric design, inclusive pricing, and governance architecture each map directly to measurable sustainability outcomes and UN Sustainable Development Goals.

Pillar	Focus Area	Core Contribution
E	Climate Adaptation Low-Carbon Operations Risk &	Parametric weather triggers for GBA climate exposure; zero-adjuster claims; digital-only distribution
S	Financial Inclusion & Protection Gap	HKD 200/month brings parametric coverage to 70% of gig workers currently uninsured
G	AI Transparency Regulatory Compliance Ethics, &	Floor-and-ceiling AI governance; full transparency dashboard; opt-in data consent; tri-jurisdictional data compliance

E — Environmental: Climate Risk Adaptation & Low-Carbon Operations

Climate adaptation through parametric triggers

Swiss Re Institute estimates Asia-Pacific accounts for the largest share of the global natural catastrophe protection gap, with insured losses covering less than 20% of economic losses from weather events in the region. Micro Protection Fluid directly addresses this gap for one of Asia's most climate-exposed urban corridors.

Low-carbon operations

The product eliminates the carbon cost of traditional insurance distribution and claims at every stage:

Feature	Environmental Impact
Digital-only underwriting	No physical forms, no branch visits, no printed policies
AI reallocation eliminates over-insurance	Reduces capital held against unused coverage; traditional home insurance continues covering an empty property while the family travels, Micro

	Protection Fluid automatically reallocates that premium to active travel risk
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SDG 13: Climate Action — Parametric triggers provide financial resilience for populations facing economic disruption across the GBA.

S — Social: Financial Inclusion & Protection Gap

The problem we are solving

The global protection gap stands at USD 9 trillion (Swiss Re Institute, "Closing the Protection Gap," 2024). 78% of insurance executives identify closing the protection gap as an ethical obligation (HSBC / EY Insurance Barometer, 2024). Micro Protection Fluid is designed from the ground up to address this gap for the segment most excluded from traditional insurance: gig workers, cross-border commuters, and informal economy participants.

Who we cover

Gig and informal workers are systematically excluded from employer-provided insurance schemes. 70% of gig workers in Asia have no accident or health insurance (ILO, World Social Protection Report, 2024). In Hong Kong, the rise of platform-economy work — delivery riders, freelancers, cross-border traders — has created a large and growing population with no coverage baseline.

Micro Protection Fluid addresses this directly: HKD 200/month brings parametric life, health, travel, liability, cyber, and property coverage within reach of workers priced out of traditional bundled policies, improving financial resilience without requiring financial sophistication.

Financial inclusion in numbers

Metric	Value	Source
Global protection gap	USD 9 trillion	Swiss Re Institute, 2024
Asian gig workers without insurance	70%	ILO, World Social Protection Report, 2024
GBA cross-border trips (H1 2025)	15.03 million	CCTV / People's Daily

Executives citing protection gap as ethical obligation	78%	HSBC / EY Insurance Barometer, 2024
Consumers wanting data transparency in dynamic pricing	78%	EY Global Insurance Consumer Survey, 2024

SDG 1: No Poverty — Micro-premium pricing brings insurance access to gig workers and informal economy participants excluded from traditional schemes.

SDG 3: Good Health and Well-being — Parametric hospital admission coverage extends health protection to the 70% of gig workers currently uninsured.

SDG 8: Decent Work and Economic Growth — Designed specifically for informal and cross-border workers excluded from employer-provided insurance schemes.

SDG 10: Reduced Inequalities — Closes the coverage gap between formally employed workers with employer insurance and informal economy participants with none.

G — Governance: AI Ethics, Transparency & Regulatory Compliance

AI governance architecture

The insurance industry's caution around AI is justified. Micro Protection Fluid is designed with this caution built in at the architecture level, not added as a disclosure afterthought.

Governance Feature	How It Works
Floor-and-ceiling model	AI cannot drop any category below its pre-approved floor (HKD 5 non-life; HKD 90 life — immovable). AI cannot create new rules autonomously.
Human oversight	All AI-suggested optimizations require human approval before implementation. No automated decision affects coverage without an actuarially certified rule governing it.
Practical AI role	AI reads MCC codes, interprets flight status APIs, flags reallocation suggestions for actuarial review, and drafts claim summaries for human approval. It does not make underwriting decisions.
Transparency dashboard	Full in-app dashboard shows customers exactly what data triggered which reallocation, in plain language. Complies with HKMA Explainable AI Guideline (2023).

Opt-in data consent	Customers explicitly consent to each data source (location, MCC, Octopus, flight status) before it is used. Consent can be withdrawn at any time.
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Cross-border data governance

Micro Protection Fluid operates across three legal jurisdictions with materially different data protection regimes. Cross-border data flows are handled through HSBC's existing data governance infrastructure, which already operates across all three. Customer data collected in each jurisdiction is processed under that jurisdiction's applicable law.

Jurisdiction	Applicable Law	Key Requirement
Hong Kong	PDPO (Cap. 486), DPP3	Explicit consent for data use in automated decisions
Mainland China	PIPL (2021)	Data localisation; cross-border transfer requires standard contract or certification
Macau	Law 8/2005	Consent-based; broadly aligned with HK PDPO

SDG 16: Peace, Justice and Strong Institutions — Transparent AI governance, multi-jurisdictional regulatory compliance, and mandatory human oversight set a standard for responsible AI deployment in financial services.

SDG Summary

SDG	Connection to Micro Protection Fluid	Evidence
SDG 1 No Poverty	Micro-premium brings insurance to gig workers excluded from traditional schemes	ILO: 70% of Asian gig workers uninsured
SDG 3 Good Health	Parametric hospital admission coverage for workers currently without health cover	78% of GBA under-35s lack adequate health coverage

SDG 8 Decent Work	Designed for informal and cross-border workers outside employer-provided insurance schemes	15.03M GBA cross-border trips H1 2025; gig economy growth
SDG 10 Reduced Inequalities	Closes coverage gap between formally employed workers with employer insurance and informal workers with none	USD 9T global protection gap (Swiss Re, 2024)
SDG 13 Climate Action	Parametric triggers provide adaptive coverage for GBA commuters	Swiss Re: <20% of Asia-Pacific weather losses insured
SDG 16 Strong Institutions	Transparent AI governance across three legal jurisdictions; human oversight; explainable rules engine	HKMA Explainable AI Guideline, 2023

Part 9: Greater Bay Area (GBA) Regional Specifics

Micro Protection Fluid is built around a single, concrete GBA cross-border scenario, not a generic insurance product with a Hong Kong label on it. Every coverage shift in the product traces back to a verifiable signal from the GBA transit and payments corridor, and every regulatory mechanism used is one that already exists for this specific corridor. This section walks through the full scenario, the data partners behind it, and why the underlying mechanics could not be replicated in a non-GBA market.

3.1 The customer: Chen Wei, 26

Wei is a UX designer who lives in Tuen Mun, Hong Kong, and commutes to a tech company in Shenzhen Bay Tech Park 3 days a week. She crosses the border roughly 140 times a year, every time via the MTR High-Speed Rail (XRL) from West Kowloon to Shenzhen Futian.

Profile	Detail
Lives in	Tuen Mun, Hong Kong

Works in	Shenzhen Bay Tech Park, 3 days/week
Income	HK\$180,000 / year
Border crossings	~140 per year (3x/week by MTR XRL)
Current cover	Basic life policy only — no travel cover, no mainland health cover

Wei has never bought travel insurance for her Shenzhen trips, they feel too short to bother with. Her existing policy gives her zero protection on the mainland. If she sprains her ankle at a Shenzhen co-working space, she is fully uninsured. Micro Protection Fluid fixes this without Wei doing anything differently: it lives inside her existing HSBC app and responds to data HSBC already sees from her banking activity.

3.2 Wei's Tuesday: how the product follows her across the border

The diagram below traces a single working Tuesday. Each trigger point corresponds to a real, existing data signal, an Octopus tap, a border-crossing confirmation, a mainland payment, and each signal causes the AI to reallocate Wei's coverage ceilings within her fixed HK\$200/month premium. Nothing about her payment changes; only the shape of her protection does.

Wei's Tuesday commute: Tuen Mun to Shenzhen Bay Tech Park

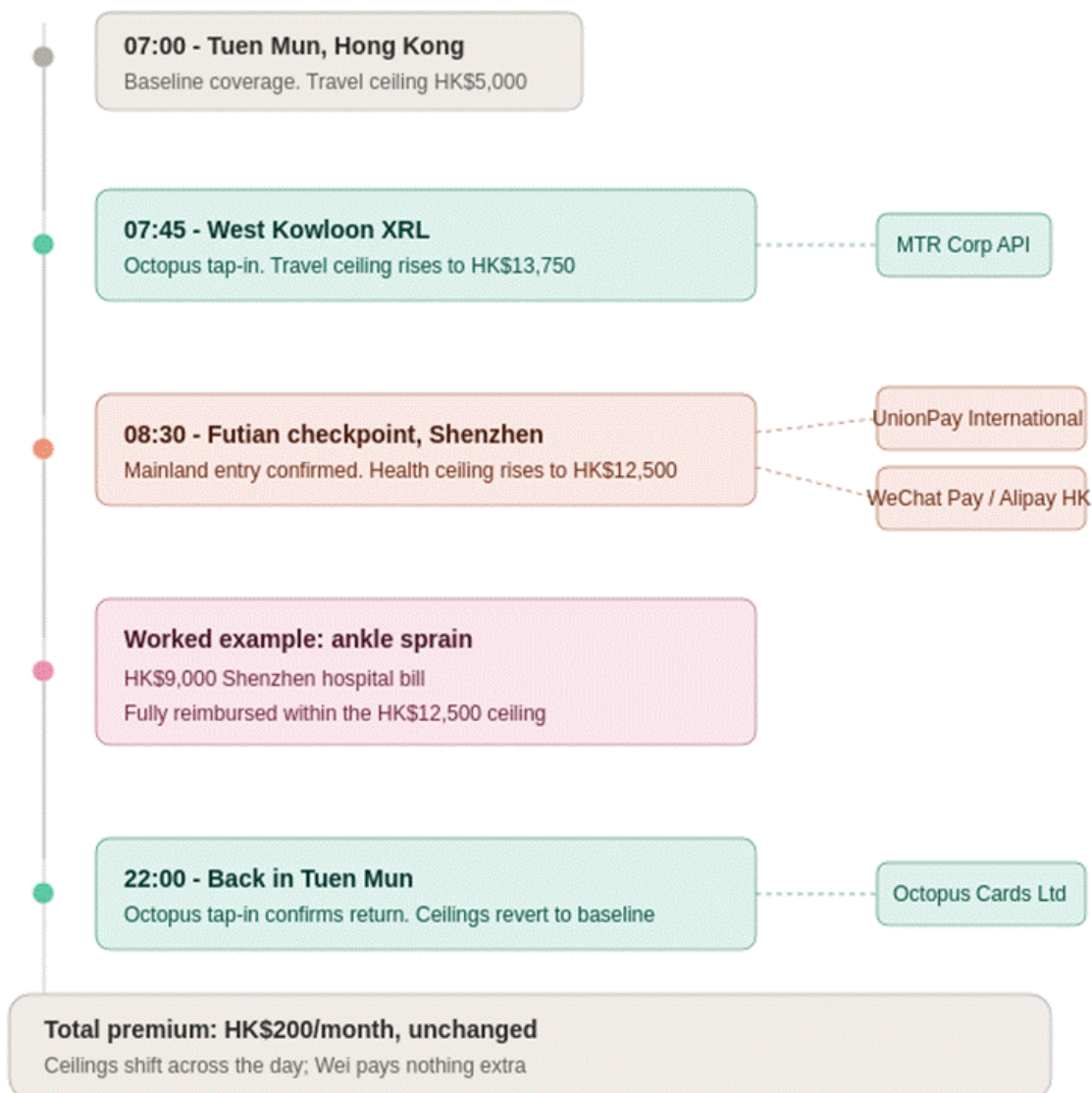


Figure 1. Wei's Tuesday commute, showing trigger points, coverage ceiling shifts, and the data partner behind each trigger.

3.3 Coverage allocation at the border

At the moment Wei crosses into the mainland (08:30, Futian checkpoint), her HK\$200 monthly allocation is redistributed across categories to reflect what she is actually exposed to that day. The table below shows her allocation while she is in Shenzhen, compared with her baseline allocation while she is in Hong Kong.

Category	Hong Kong baseline / Shenzhen allocation
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Travel	HK\$5,000 baseline → HK\$13,750 ceiling (55% of allocation) once XRL boarding is detected
Health	Standard HK cover → HK\$12,500 mainland health ceiling (25%) once mainland entry is confirmed
Life	HK\$2,500 ceiling (10%) — unchanged
Cyber	HK\$1,500 ceiling (6%) — unchanged
Property / liability	HK\$1,000 combined ceiling (4%) — unchanged
Total premium	HK\$200/month — fixed, never changes

3.4 Worked example: the ankle sprain

Scenario: Wei sprains her ankle at a co-working space in Shenzhen Bay Tech Park. Hospital bill: HK\$9,000.

- Wei pays the Shenzhen hospital HK\$9,000 upfront, as she would for any out-of-network mainland facility.
- She uploads the receipt via the HSBC app once back in Hong Kong — no separate claims portal, no adjuster visit.
- HSBC reimburses the full HK\$9,000. Her health ceiling at the time of the claim was HK\$12,500, so the entire bill falls within it.

Without Micro Protection Fluid: Wei's existing basic life policy pays HK\$0 for a physical injury sustained on the mainland. She would be fully out of pocket for the HK\$9,000.

3.5 Why this could not be a generic product

- Dual-jurisdiction health shield — When Wei crosses into Shenzhen, a separate health layer activates, sized for mainland hospital costs and reimbursed in HKD by HSBC. She does not need a separate China health policy. This works through the Wealth Management Connect (WMC) scheme — the Hong Kong–mainland China government framework that lets HSBC offer a limited set of cross-border financial products to GBA residents without obtaining full CBIRC (China Banking and Insurance Regulatory Commission) approval from scratch. Normally, paying out an insurance claim in CNY to a customer in Shenzhen would require heavy mainland regulatory clearance; WMC provides an existing legal corridor for exactly this kind of limited cross-border payout. No equivalent sandbox exists in any other market HSBC operates in.

- **Activity-based dynamic allocation tailored to the commute corridor** — The AI shifts coverage ceilings across categories based on real, verifiable signals specific to the GBA transit corridor itself: MTR XRL boarding via Octopus, border-crossing confirmation via mainland payment activity, and return-journey detection. Protection matches what Wei is actually exposed to at each point in her day, while the fixed HK\$200/month premium never changes. This is a product mechanic that only makes sense for a population making repeated, predictable cross-border movements — exactly the profile of a GBA commuter.
- **No new data sharing required** — HSBC already sees Wei's Octopus taps, UnionPay transactions, and MTR journeys through its existing banking relationship with her. The product does not ask her to connect new accounts or share data with any third party. The data infrastructure was already there — Micro Protection Fluid simply activates it.

3.6 Regional data infrastructure partners

The product flow below groups the data partners that feed Wei's allocation engine into two categories: transit and border signals, and payment signals. Both feed continuously into the HSBC allocation engine, which adjusts Wei's coverage ceilings in real time.

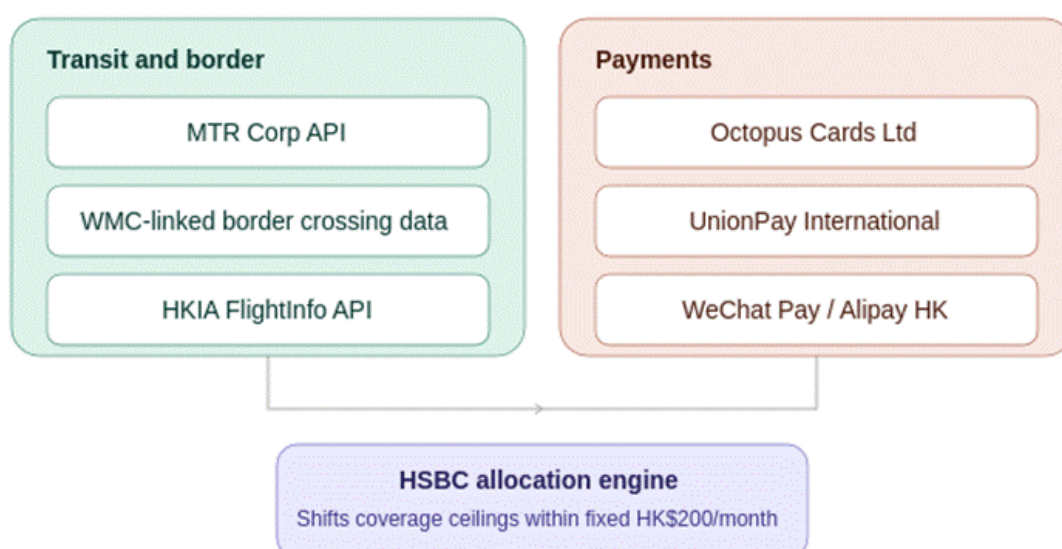


Figure 2. Regional data infrastructure partners feeding the HSBC allocation engine.

Partner	Role in Wei's journey
MTR Corp API	Detects Wei's Octopus tap at West Kowloon XRL at 07:45. This single signal tells the AI she has started

		a cross-border trip and triggers the travel ceiling increase to HK\$13,750.
WMC-linked crossing data	border	Confirms mainland entry at Futian via the Wealth Management Connect-linked transaction feeds. Triggers the mainland health ceiling top-up to HK\$12,500 immediately on confirmation.
HKIA FlightInfo API		Used on days Wei flies instead of taking the XRL — pre-activates travel coverage before she reaches the airport.
Octopus Cards Ltd		Sees her tap back into HK MTR at 22:00 — this is how the AI knows she is home and reverts all ceilings to baseline.
UnionPay International		Every mainland transaction Wei makes carries a merchant category code. The AI reads these to understand what she is doing in Shenzhen and refine the day's allocation.
WeChat Pay / Alipay HK		Fills in activity signals that card transactions miss — small purchases, street food, local transport — giving a fuller picture of her day on the mainland.

3.7 Summary: the GBA fit in one paragraph

Micro Protection Fluid does not bolt a Hong Kong wrapper onto a generic composite policy. Every part of the GBA scenario, the dual-jurisdiction health shield, the activity-based reallocation, and the data partners themselves, is specific to the reality of a resident who lives in Hong Kong, works in Shenzhen, and crosses the border on the XRL several times a week. The regulatory mechanism (WMC) is GBA-specific, the transit signals (MTR Octopus, border crossing data) are GBA-specific, and the payment signals (UnionPay, WeChat Pay/Alipay HK) reflect the dual-currency, dual-app reality of life in the corridor. This is the product friction that a GBA commuter actually faces, solved with infrastructure that already exists — which is exactly the standard an HSBC-sponsored, Hong Kong-based competition should be looking for.